

EE 622: Microwave Integrated Circuits

Course Code	EE 622
Course Name	Microwave Integrated Circuits
L-T-P-C	3-0-0-3
Prerequisites	EE507 - Transmission Lines and Basic Microwave Engineering or equivalent, PH52 I - Electromagnetic Theory or equivalent
Intended for	BTech Third and Final year/M.Tech./MS/PhD
Distribution	Elective for third and final year Electrical Engineering, MTech VLSI, And Communication and Signal Processing (CSP), M.S., Ph.D.
Approval	18th BoA (Sep 2017); 14th Senate (Oct 2017)

Course Contents

- **Introduction to Printed Transmission lines and Transitions:** Concept of 2 Port Network and S Parameters, Matching transmission line sections and theory of multiple reflections, Transitions in different transmission lines, Micro-Strip and Strip lines, Left handed transmission lines. Smith chart: Double stub Matching. [8 Lectures]
- **Power Dividers and Directional Couplers:** Basic Properties of power dividers and couplers, T junctions, Wilkinson type, quadrature hybrid power dividers, coupled line directional coupler, 90 deg. and 180 deg. - Hybrid branch line couplers. Circulators-Active and Passive. Broad-Band techniques. [6 Lectures]
- **Filters and Isolator:** Basic Filter design techniques like image parameter and insertion loss, Filter transformations and implementations, low pass filters, coupled line filters, coupled resonator based filters, metamaterial filters. Ferrite isolator circuits and design. [6 Lectures]
- **Microwave Amplifiers:** Single Stage Transistor amplifier, Power gain equations, Stability circles, Broad-band amplifier design, Solid state Power amplifiers. [8 Lectures]
- **Microwave diodes and transistors:** BJT, GaAs FETs, and their applications, IMPATT, TRAPATT, Gunn diodes. [4 Lectures]
- **Oscillators and Phase Shifters:** Microwave oscillators using Transistors, dielectric resonators, Active and passive Phase-Shifters. [6 Lectures]
- **Technologies in MIC:** Monolithic and Hybrid substrates, Metamaterial substrates, Millimeter wave techniques, Simulations based study of printed microwave device. [4 Lectures]

Note: All the Units will have tutorials and in the end of course there will be a presentation assignment in which student (or a group) will present latest advancement in any-one of the topics taught in class

Text Books:

1. David M Pozar, **Microwave Engineering**, Fourth Edition, John Wiley & Sons Publications.
2. G. Gonzalez, **Microwave transistor amplifier: design and analysis Handbook**, 2nd Edition, Prentice hall Publications.

References

1. Leo Young and H. Sobol, Ed., **Advances in Microwaves**, vol.2, Academic press.Inc.
2. S. Y. Liao, **Microwave circuit and analysis and amplifiers design**, Prentice Hall.
3. C. Caloz and T. Itho, **Electromagnetic Metamaterial: Transmission Line Theory and Microwave Applications**, Wileys Publications.
4. Research Papers as instructed by course Instructors.